

UNDERSTANDING DIABETES RISK USING HEALTH AND LIFESTYLE INDICATORS

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IS 327

Project Overview

Dataset: CDC Diabetes Health Indicators (UCI ML Repository)

- **253,689 rows, 21 features, binary diabetes outcome**
- **Class imbalance: 86% non-diabetic, 14% diabetic**

Research Questions:

- 1. Which ML model performs best at predicting diabetes risk?**
- 2. How do lifestyle/health factors compare to socioeconomic factors?**

Models tested: Logistic Regression, Decision Tree, Random Forest, K-NN

Methods: 5-fold stratified cross-validation, class weighting for imbalance, scikit-learn pipelines, and StandardScaler

Results & Model Comparison

Model	F1	Recall	ROC-AUC
Logistic Regression	0.445	0.773	0.826
Random Forest	0.214	0.136	0.806
K-NN	0.226	0.200	0.717
Decision Tree	0.289	0.289	0.588

Results: Lifestyle vs Socioeconomic

Random Forest Importance:

- Lifestyle features: 68.5%
- Socioeconomic features: 31.5%on-diabetic

Top 5 predictors: BMI, Age, General Health, Income, High Blood Pressure

Performance by feature set (Logistic Regression):

- Lifestyle only F1 = 0.435
- Socioeconomic only: F1 = 0.338
- Combined: F1 = 0.445

Takeaways: Lifestyle features dominated, but combining both gave the best result.

Conclusions & What I learned

Hypotheses revisited:

- Random forest performs best x (logistic regression won)
- Lifestyle > socioeconomic
- Combined performed best

Biggest lesson: Accuracy is misleading on imbalanced data. Recall and F1 told a much clearer story

Why this matters: Picking the right features and metrics are just as important as picking the right model. Especially in a healthcare setting.